

RAMCO INSTITUTE OF TECHNOLOGY

Department of Computer Science and Engineering

Academic Year: 2019 - 2020 (Odd Semester)

Degree, Semester & Branch: VII Semester B.E. Computer Science and Engineering.

Course Code & Title: CS6703 Grid and Cloud Computing

Name of the Faculty member: M.Gomathy Nayagam, Asst.Professor (SG)/CSE

Innovative practice: Flipped Class

Type of Learning: Collaborative Learning

Topic: HDFS Concepts

Date: 16.09.2019

Learning Objectives:

Objective	Statements
O1	Develop the web services using the grid and cloud technologies.
O2	Students will able to do self-paced learning.
O3	To make the students to coordinate with their team to share their view of the HDFS concepts that was learned from the video and share in front of the class.

Justification for the Topic:

Apache Hadoop is one of the big data analytics tools which will be used by the most of the industry. It is highly reliable, scalable, distributed processing tool for big data. HDFS is one of component in Apache hadoop. The primary objective of the flipped class is to allow the students for self-paced learning, dive deeper into the concept, enhance the communiqué skill, inventiveness and analytic skill. Hence this activity will help the students to explore knowledge in HDFS concepts well and good.

Use of Flipped Class:

- It helps students to allow self-paced learning.
- It motivates the students to dive deeper into HDFS concepts
- It helps the students to be better prepared on the particular topic.

Procedure of this Flipped Class:

- It is an instructional approach and a type of blended learning environment by delivering instructional content, online, outside of the class room.
- A video lecture hosted on online one week before of the class discussion.

- Students are instructed to watch the video from their home and took the notes individually. They asked to prepare and come for discussion in the class room on particular day.
- This will help the students to improve their self-learning and better preparation on the particular topic.

Activity:

- Video lecture corresponding to the particular topic was hosted in online one week before.
- Students were instructed to watch the video and took the notes according to their understanding level in their home and come with preparation for discussion on one particular day in the class room.
- The students were segmented as a group in the class during the day of discussion.
- We gave 10 minutes for discussion among the group member regarding the video lecture they watched earlier.
- Each group collected all the important points related to the topic from those group members and to be allotted 15 minutes time for on board presentation and discussion about the topic in front of the whole class with their corresponding group members.
- Corresponding subject faculty monitored each group's presentation.
- Course Instructor selected one among the student in the class and asked the particular student to summarize all groups' presentation for next 5 minutes.



Fig 1: Group members are discussed about the topic among themselves.



Fig 2: Group 'A' Present their Points about the concepts of HDFS



Fig 3: Group 'C' Present their Points about the concepts of HDFS

Outcome

The students can able to explore the knowledge in HDFS concepts by improving their self-paced learning and get more key points from their peer groups when they are doing collaborative discussion among them.

Objective and PO Mapping

Objective(s)	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
O1	3		3		2							
O2	3	3	1	1	2	1	1		2	2		3
O3	3	3	1	1	1			1	2	2		2

Reflective critique

Pre-implementation Reflection

Challenges:

- Difficult to make the individual to watch the video which is already hosted in canvas.
- Difficult to Select the video for the activity and create the group which contains heterogeneous students.
- Space for conducting this activity inside the class room is not so good.

Steps taken for address the challenges

- Motivated the students to watch the video and send alert message frequently to the students through the canvas.
- Make the group of 7 members which contains toppers, above average, average and below average students.
- Conducted the activity in seminar hall which will avoid the disturbance to the neighbouring class.

Post implementation reflection.

- Students are eagerly participated in this activity.
- They felt, they learned the concept with their peer group in enjoyable manner.
- Students gave the feedback that they understood the HDFS concept well and good

References

1. <https://facultyinnovate.utexas.edu/flipped-classroom>
2. <https://omerad.msu.edu/teaching/teaching-strategies/27-teaching/162-what-why-and-how-to-implement-a-flipped-classroom-model>
3. <https://www.heacademy.ac.uk/knowledge-hub/flipped-learning-0>

Faculty in-charge

HOD/CSE